PROJECT BASED LEARNING (PBL-4) LAB (CSP390)

AI-Powered Taskade-Like Productivity System

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Project Title

AI-Powered Taskade-Like Productivity System:- The AI-Powered Taskade-Like Productivity System is a unified platform that integrates task management, communication, and note-taking features. It uses artificial intelligence to provide smart recommendations such as task prioritization and deadline prediction. The system improves collaboration, reduces dependency on multiple tools, and enhances overall productivity and workflow efficiency for users**.**

Team / Group Formation:

Table forgroupmembers-1

S. No	Student Name	Roll Number	System ID	Role
1	Preeti Gupta	2301010649	2023354258	Developer/Debugger
2	Shubham Kumar Mandal	2301010837	2023499588	Backend Developer
3	Tanu Mishra	2301010900	2023428136	Frontend Developer

Technologies to be used

Software Platform

a) Front-end:

- React.js(Version 18+) – for building interactive user interfaces
- TailwindCSS / Material UI (MUI) – for responsive and modern UI design
- HTML5,CSS3, JavaScript (ES6+) – for structuring and styling web pages

b) Back-end:

- Node.js(Version 18+) – runtime environment for server-side development
- Express.js(Version 4+) – web framework for building REST APIs

- FastAPI – for AI-based microservices

c) Database:

- MongoDB (Version 6+) – NoSQL database for storing application data

d) APIs & Integration:

- Socket.io – for real-time communication (chat, notifications)
- OpenAI API / Gemini API (optional) – for AI-powered task suggestions
- JWT (JSON Web Token) – for authentication and security

Hardware Platform

- **RAM:** Minimum 8 GB (16 GB recommended)
- **HardDisk:** Minimum 256 GB SSD
- **Processor:** AMD Ryzen 5 or above
- **OperatingSystem:** Windows 10/11
- **Editor/IDE:** Visual Studio Code (VS Code)
- **Browser:** Google Chrome / Microsoft Edge

Tools

Table for list of tools-2

Tool Name	Vendor	Version	Purpose
Visual Studio Code	Microsoft	Latest	Code editing and development
Git & GitHub	GitHub Inc.	Latest	Version control and collaboration
Postman	Postman Inc.	Latest	API testing and debugging
Docker	Docker Inc.	Latest	Containerization and Deployment
MongoDB Compass	MongoDB Inc.	Latest	Database visualization and management

Problem Statement

In today's digital world, managing tasks efficiently is essential, but existing tools like task managers, communication apps, and note-taking platforms often work separately, causing poor integration. This forces users to switch between apps, leading to wasted time, scattered information, missed deadlines, and reduced productivity. Additionally, most systems lack intelligent features, requiring users to manually prioritize tasks and manage schedules, which becomes difficult as workload increases.

This fragmented approach also creates challenges in teamwork, where lack of coordination and transparency can slow down progress. Users often struggle to track overall performance and adapt to changing priorities. The main problem is the absence of a unified and intelligent productivity system. The proposed AI-powered platform addresses this by integrating key functionalities into one system and using AI to automate prioritization, provide smart insights, and improve decision-making, ultimately enhancing productivity, collaboration, and workflow efficiency.

Moreover, such a system can provide real-time updates, better data organization, and personalized recommendations based on user behavior. This helps individuals and teams stay focused, reduce stress, and maintain consistency in their work, ensuring more effective task execution and improved overall performance.

Literature Survey

The advancement of Artificial Intelligence (AI) has significantly transformed productivity tools and task management systems. Researchers have explored the role of AI in improving efficiency, automating repetitive tasks, and supporting better decision-making. This section reviews relevant studies related to AI-based productivity systems and collaborative platforms.Recent studies indicate that AI-powered tools enhance productivity by utilizing technologies such as machine learning, natural language processing, and predictive analytics. These technologies help users organize tasks, analyze large amounts of data, and improve workflow efficiency. AI-based systems are increasingly being adopted in various domains, including education, research, and business environments, due to their ability to reduce manual effort and increase accuracy (Litmaps, 2024).A study conducted on AI productivity tools in research environments highlights that users generally find these tools beneficial for improving efficiency and simplifying complex tasks. However, the study also identifies challenges such as lack of trust, high implementation cost, and concerns related to data reliability and plagiarism. These issues indicate that while AI tools offer advantages, their adoption depends on user confidence and proper implementation (Ponciano et al., 2025).

Research focusing on organizational productivity suggests that AI integration plays a vital role in enhancing workplace performance. AI systems assist in automating routine processes, improving decision-making, and enabling organizations to respond effectively to changing demands. This leads to increased efficiency and better utilization of resources (Kassa, 2025). Similarly, other studies emphasize that AI-based tools help streamline operations and improve overall workflow management in professional environments (IJSRET, 2025).In the domain of task management, AI-based systems have been developed to provide intelligent task prioritization. These systems analyze user behavior, deadlines, and task patterns to suggest optimal workflows. Such approaches improve time management and help users focus on high-priority tasks, thereby increasing productivity (IJERT, 2024).Furthermore, research on AI-assisted development tools demonstrates that AI can significantly enhance user performance. Experimental studies show that AI-based assistants enable users to complete tasks more quickly by automating repetitive processes and providing real-time suggestions. However, some limitations such as over-dependence on AI and the need for user training have also been identified (Peng et al., 2023).Overall, the reviewed literature highlights that AI-powered productivity systems have strong potential to improve efficiency, reduce workload, and enhance collaboration. At the same time, challenges such as trust, ethical concerns, and proper implementation need to be addressed. These findings support the development of an integrated AI-based productivity system that combines task management, communication, and intelligent assistance into a single platform.

Table for Literature Survey-3

Author(s) / Year	**Title of Study**	**Methodology**	**Key Findings**
Litmaps (2024)	AI Research Tools Overview	Conceptual Study	AI improves efficiency using ML and NLP techniques
Ponciano et al. (2025)	AI Productivity Tools in Research Writing	Survey & Analysis	Identified benefits along with trust and cost issues
Kassa (2025)	Impact of AI on Organizational Performance	Analytical Study	AI enhances productivity and decision-making
IJSRET (2025)	AI Tools in Workplace Productivity	Review Study	AI streamlines operations and workflow
Peng et al. (2023)	AI in Developer Productivity	Experimental Study	AI reduces task completion time
IJERT (2024)	Task Management System using AI	System Design	AI enables smart task prioritization

Project Description

Theproposed project, *AI-PoweredTaskade-LikeProductivitySystem*, is designed to provide a unified platform for managing tasks, communication, and notes with the support of Artificial Intelligence. The scope of this project includes developing a web-based application that enables users to efficiently organize their work, collaborate with team members, and receive intelligent recommendations for improving productivity.The system aims to eliminate the need for multiple standalone applications by integrating all essential productivity features into a single interface. Users can create and manage tasks, communicate in real time, maintain notes, and track progress within one platform. The inclusion of AI enhances the system by providing smart suggestions such as task prioritization, deadline estimation, and workflow optimization.The structure of the project is modular, ensuring scalability and easy maintenance. The system consists of different interconnected components such as user interface, backend server, database, and AI engine. Each component performs a specific function while contributing to the overall system performance.

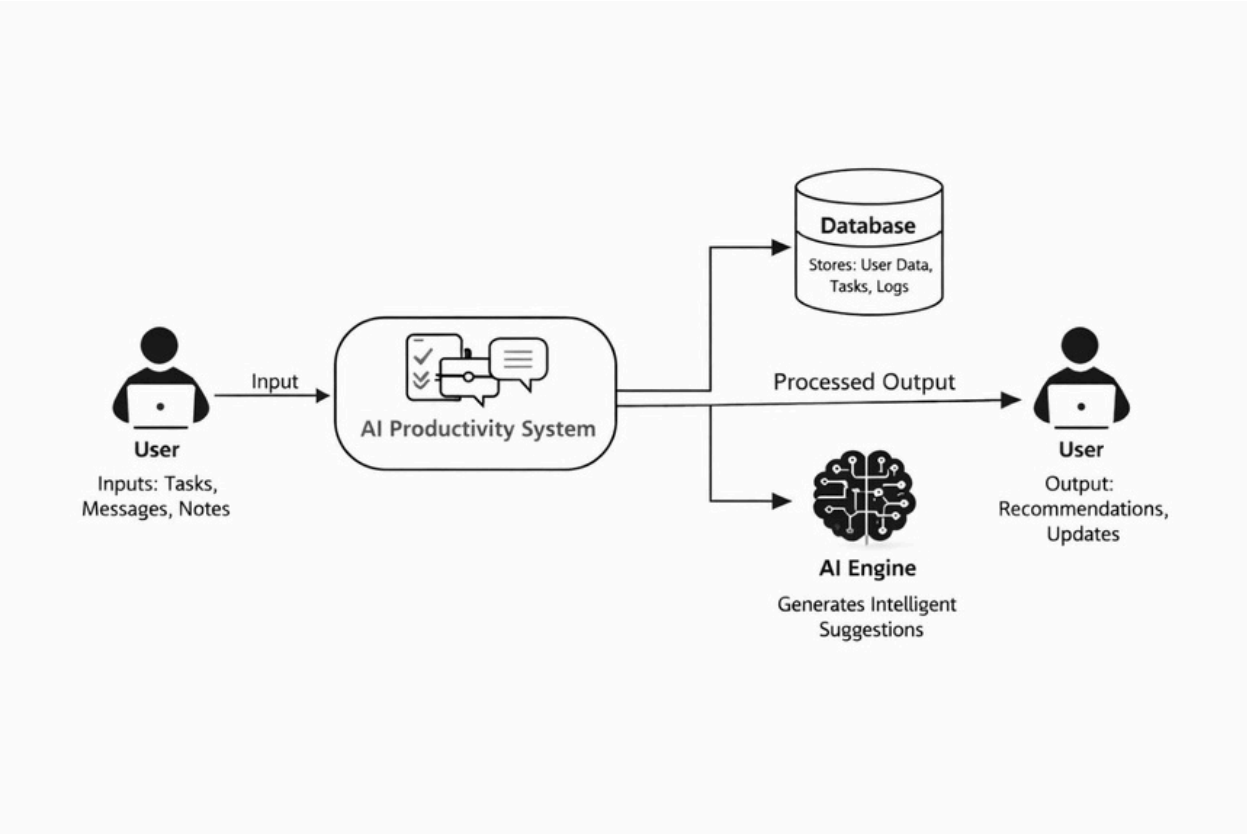

Fig-1 Workflow Representation of AI-Based Task Management System

Project Modules: Design/Algorithm

The proposed system is dividedintothefollowing modules:

1. User Authentication Module
The User Authentication Moduleisresponsible for handling user registration and login processes. It ensures secure accesstothesystem by using authentication mechanisms such as JSON Web Tokens (JWT). Thismodulealso manages user sessions, allowing users to access the platform securely without repeatedlogins.

2.Task Management Module
The Task Management Moduleenablesusers to create, update, delete, and assign tasks. It supports features such as taskcategorization, deadline management, and status tracking. This module helps users organizetheirworkefficiently and monitor progress effectively.

3. Communication Module
The Communication Moduleallowsreal-time messaging between users, supporting team collaboration and discussions.Itusestechnologies such as WebSockets (Socket.io) to ensure fast and reliable communicationwithinthesystem.

4. Notes Management Module
The Notes Management Moduleprovidesfunctionality for creating, editing, and organizing notes. It supports formatting andstorageofinformation, helping users document ideas and maintain structured records.

5.AIRecommendationModule
The AI RecommendationModuleanalysis user activity, task history, and deadlines to generate intelligent suggestions.Itprovidesfeatures such as task prioritization, deadline prediction, and workflow improvementusing machine learning or rule-based algorithms.

6. Notification Module
The Notification Modulekeeps users updated by sending reminders for deadlines, alerts for task changes, and notificationsfor messages. This ensures users remain informed and can manage their tasks effectively.

7.DatabaseManagement Module
The Database Management Moduleis responsible for storing and managing all system data, including user details, tasks, messages, and notes. It ensures data consistency, security, and reliable access to information.

Implementation Methodology

The development of the proposed *AI-Powered Taskade-Like Productivity System* follows a structured and systematic approach to ensure efficiency, scalability, and reliability. The methodology adopted for this project is based on the Agile development model, which allows iterative development, continuous feedback, and flexibility in incorporating changes during the development process.

1. Requirement Analysis
The development processbegins with identifying user needs and system requirements. This includes defining core functionalities such as task management, communication, note-taking, and AI-based recommendations. The project scope and objectives are clearly established in this phase.

2. System Design
In this phase, theoverall architecture of the system is planned. Various diagrams are used to represent systemstructure and data flow. A Data Flow Diagram (DFD) illustrates how data moves betweenusers, the system, the database, and the AI module. An Entity Relationship (ER) Diagram definesrelationships between entities such as users, tasks, messages, and notes. A Use Case Diagramrepresents user interactions, while a Class Diagram shows system structure. Flowcharts areused to explain process flow.

3. Implementation (Development Phase)
The system is developed using modern technologies. The frontend is built using React.js for an interactive interface. The backend is developed using Node.js and Express.js to handle business logic and APIs. MongoDB is used for data storage. The AI module is implemented using machine learningor rule-based algorithms to provide intelligent recommendations.

4.TestingPhase
The system is tested to ensurereliability and performance. Unit testing is performed on individual modules, integrationtesting ensures proper interaction between components, and system testing validates overallfunctionality. User Acceptance Testing (UAT) is conducted to confirm that the system meetsuser requirements.

5. Defect Log Maintenance
All identified bugs and issuesare recorded in a defect log. Each entry includes details such as issue description, severity, status, and resolution. This helps in tracking and managing errors efficiently.

6.Deployment
After successfultesting, the system is deployed for user access. The application is hosted on a server, makingitavailable for real-time use.

7. Maintenance and Updates

Post-deployment, thesystemisregularly maintained. Bug fixes, performance improvements, and feature updates are implementedbased on user feedback to ensure long-term usability and efficiency.

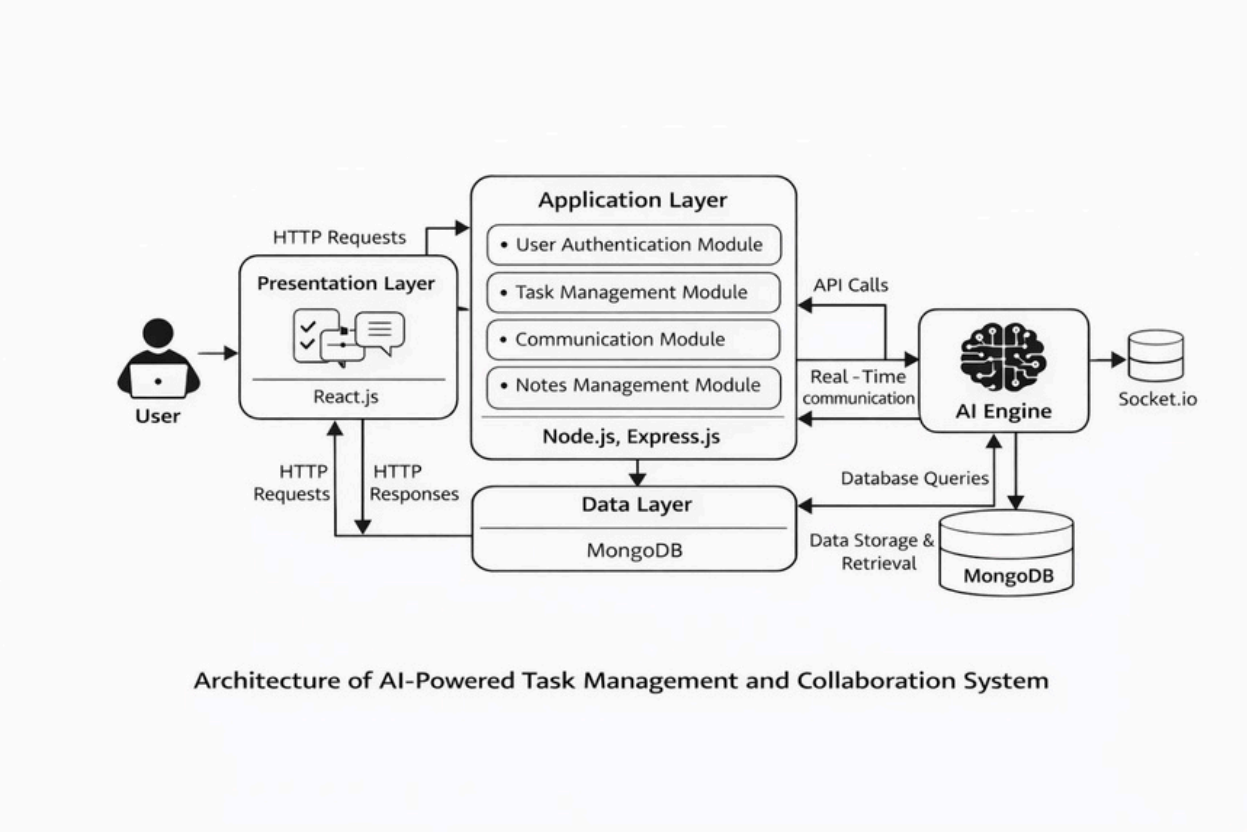

Fig.2

Result & Conclusion

The developed *AI-Powered Taskade-Like Productivity System* successfully integrates multiple productivity features such as task management, real-time communication, and note-taking into a single unified platform. The system provides a user-friendly interface that enables users to efficiently create, manage, and track tasks while collaborating with team members in real time. The use of technologies like React.js, Node.js, and MongoDB ensures smooth performance, scalability, and efficient data handling.

The implementation of the AI module enhances the overall functionality by providing intelligent suggestions, including task prioritization, deadline estimation, and workflow improvements. These features help users make better decisions, reduce manual effort, and manage their workload more effectively. Additionally, real-time communication through Socket.io improves coordination and ensures seamless interaction among users.

The system performs efficiently under different conditions, handling multiple users and maintaining data consistency and security. Testing results confirm that all modules function correctly, and the system is reliable, responsive, and easy to use. It effectively reduces the need for multiple separate tools, thereby simplifying workflow and improving productivity.

In conclusion, the proposed system successfully addresses the limitations of traditional task management tools by offering an integrated and intelligent solution. The combination of essential productivity features with AI-driven capabilities enhances user experience and supports better decision-making. The modular design ensures flexibility and scalability, making the system suitable for both individual users and teams. Overall, the project demonstrates the effectiveness of using Artificial Intelligence in building a smart productivity platform that improves efficiency, collaboration, and overall work management.

Future Scope and further enhancement of the Project

1. Advanced AI Integration
Thesystem can beenhancedby integrating advanced AI techniques such as deep learning and natural language processing.This will enable more accurate task recommendations, automated scheduling, and features likevoice-based commands and virtual assistants.

2. Mobile Application Development
Developing a mobileapplication forAndroid and iOS will allow users to access the system anytime and anywhere. This will improve accessibility and provide a better user experience.

3. Advanced Analytics and Reporting
The systemcaninclude analyticaldashboards to track productivity, task completion rates, and time management. These insightswill help users improve their workflow and decision-making.

4. Enhanced Security Features
Securitycanbeimproved byimplementing multi-factor authentication, data encryption, and role-based access control toprotectuser data and ensure safe system usage.

5. Third-Party Integration
Integrationwithexternaltoolssuch as email services, calendars, and cloud storage can make the system more connectedandefficient for users.

6. Offline Functionality
Anofflinemode canbeaddedto allow users to work without an internet connection and synchronize data oncethey areonline.

7. Multi-Language Support
Adding supportformultiplelanguages will make the system accessible to a wider range of users globally.

8. Scalability for Large Organizations
The systemcanbe expandedtosupportlarge teams and organizations by adding features like team management, project tracking, andperformance evaluation.

9. Voice Assistant Integration
Voice-basedcommandscanbeadded to allow users to create tasks and manage activities using speech.

10. Smart Notifications and Reminders
Thenotificationsystemcanbeimprovedto provide context-aware and priority-based alerts.

11. Integration of Calendar Scheduling
Abuilt-insmartcalendarcanhelpusersplan schedules, set deadlines, and avoid conflicts.

Advantages of this Project

1. Unified Platform
Thesystem combinestask management, communication, and note-taking into a single platform, reducing the need formultiple applications.

2. Improved Productivity
Byorganizingtasksandproviding a structured workflow, the system helps users complete work more efficiently.
3. AI-Based Smart Suggestions
Theintegration ofArtificialIntelligence enables features like task prioritization, deadline prediction, and workflow optimization.
4. Real-Time Collaboration
Userscancommunicateinstantly through real-time messaging, improving teamwork and coordination.
5. User-Friendly Interface
Thesystem providesaneasy-to-use interface, making it accessible for both beginners and experienced users.

6. Time Saving
Byreducingmanual effort and automating processes, the system saves time and increases efficiency.
7. Better Task Organization
Userscancategorize,track,and manage tasks effectively, leading to improved work management.

8. Scalability
Thesystemcan bescaled to support individuals, teams, and large organizations.

9. Data Security
Secureauthentication anddata handling mechanisms ensure user information is protected.

10. Reduced Workload
Automationandintelligentrecommendations reduce the burden of manual decision-making.

11. Enhanced Decision Making
AIinsightshelpusersmakebetterdecisions regarding task prioritization and scheduling.

12. Centralized Data Management
Alldataisstoredinoneplace,making it easy to access, manage, and retrieve information

Outcome

Project to Product-

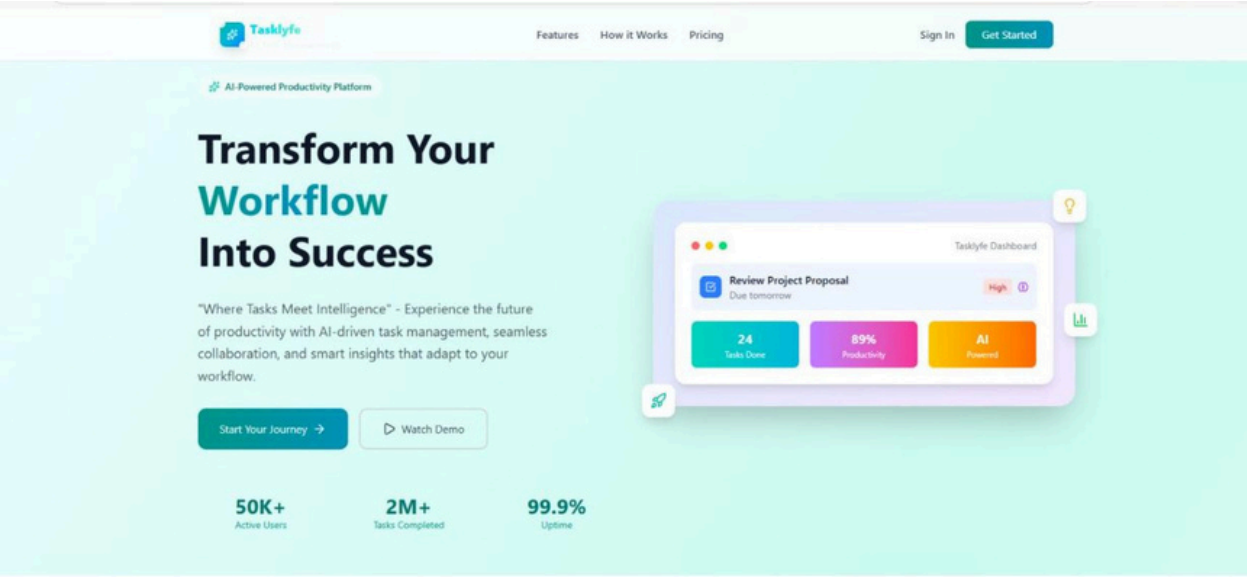

Fig-3 Landing Page

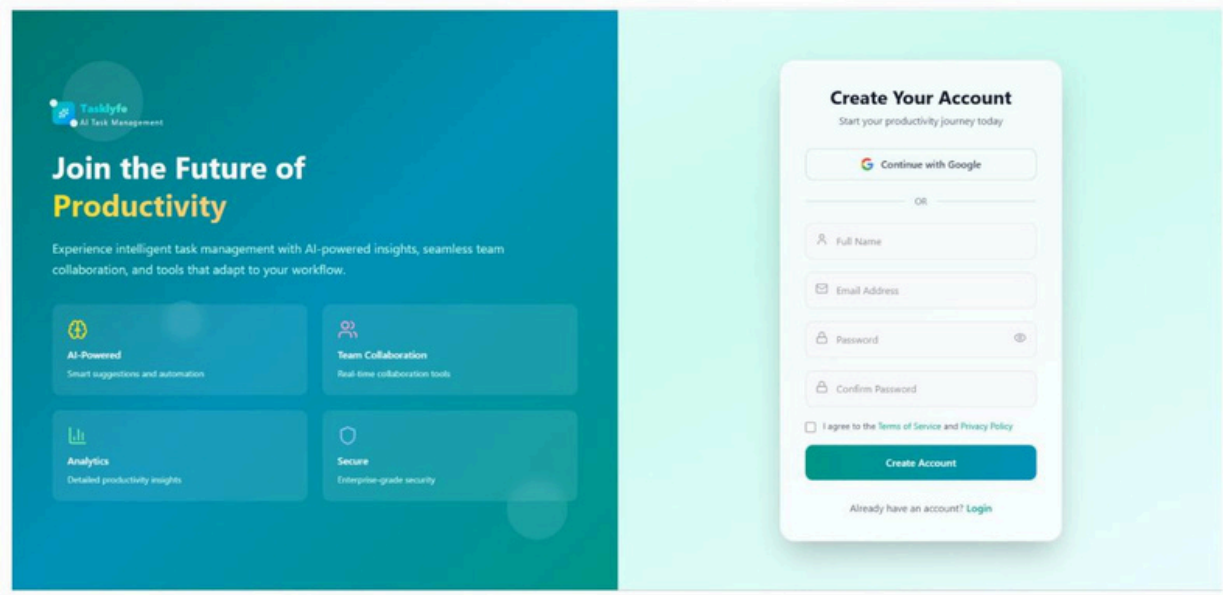

Fig-4 Sign In Page

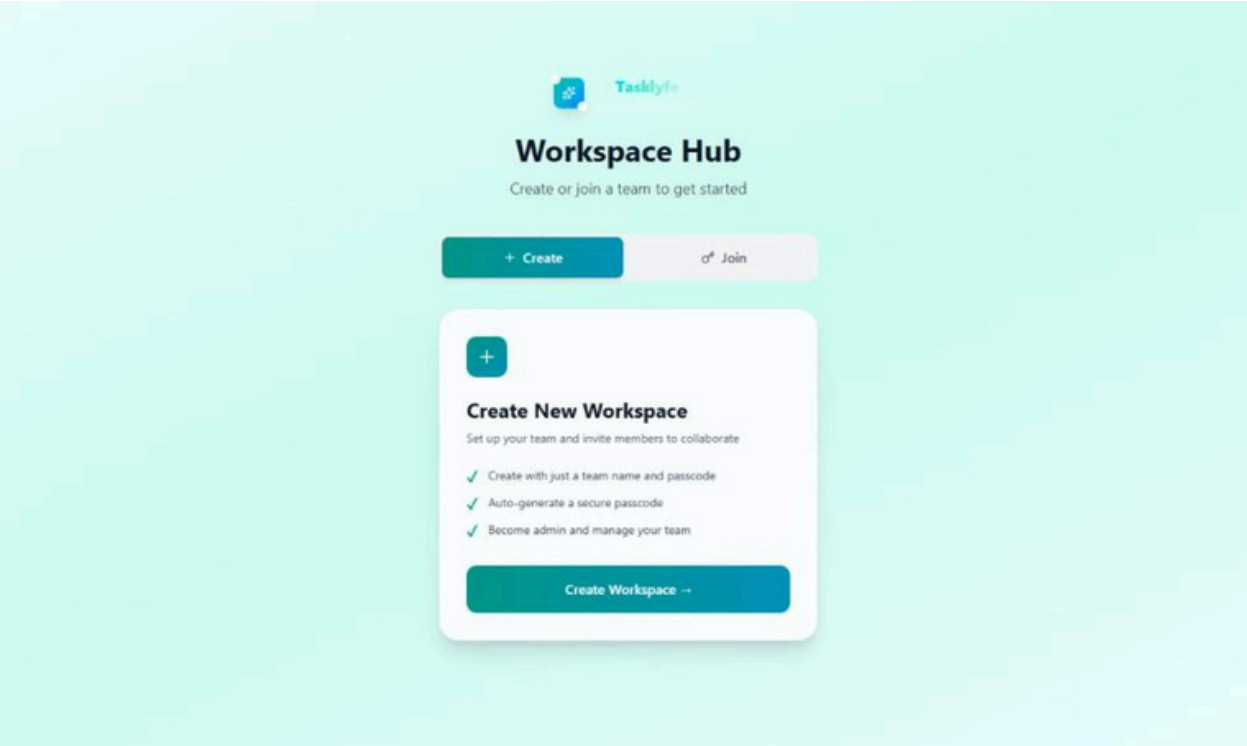

Fig-5 Work Space

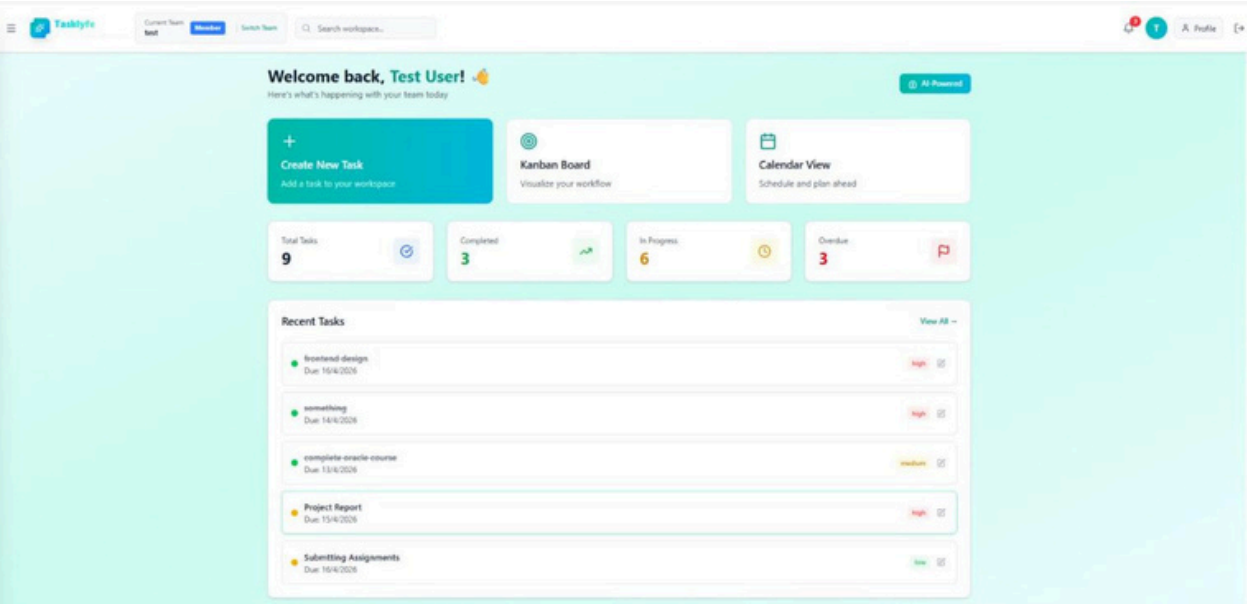

Fig-6 User DashBoard

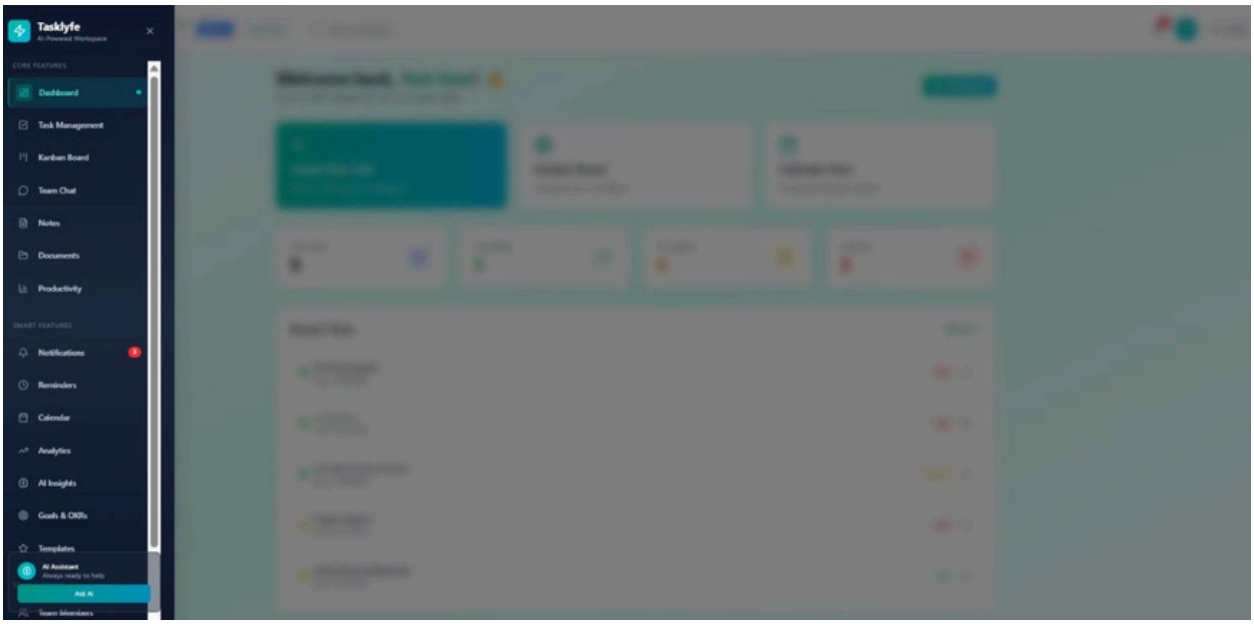

Fig-7 SideBar

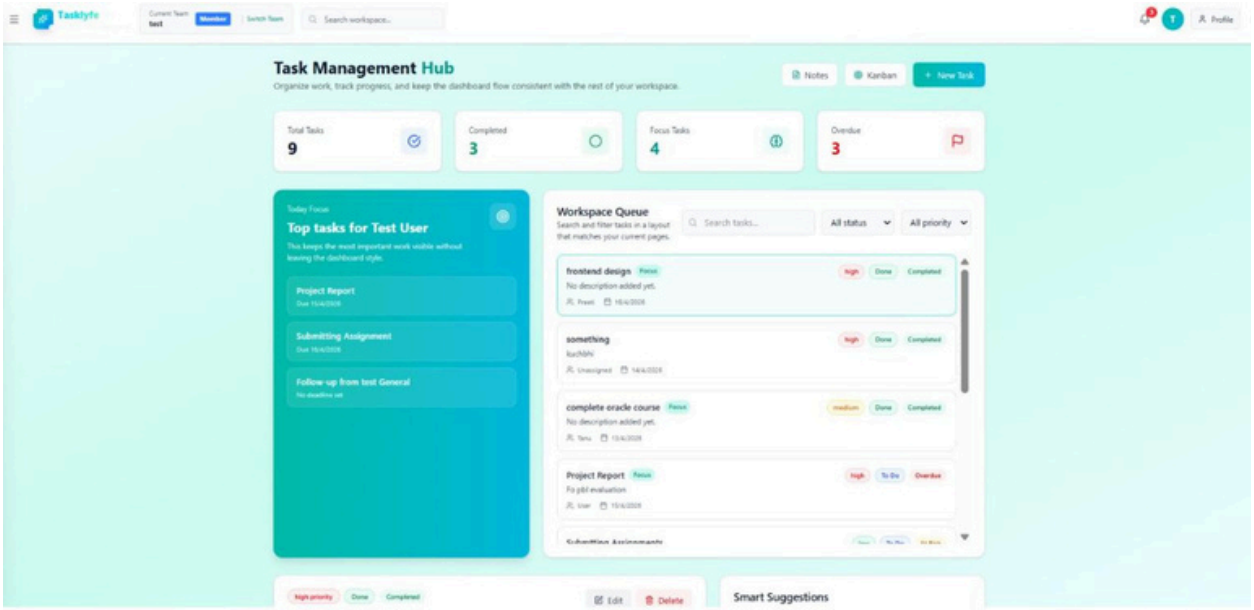

Fig-8 Task Management Hub

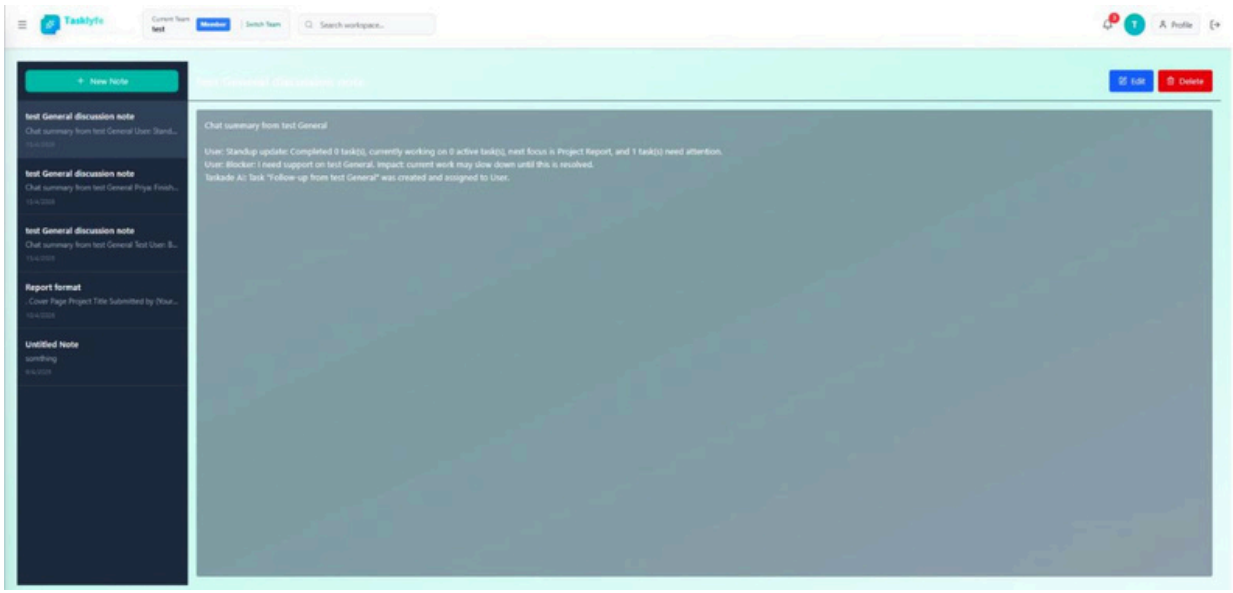

Fig-9 NotePad

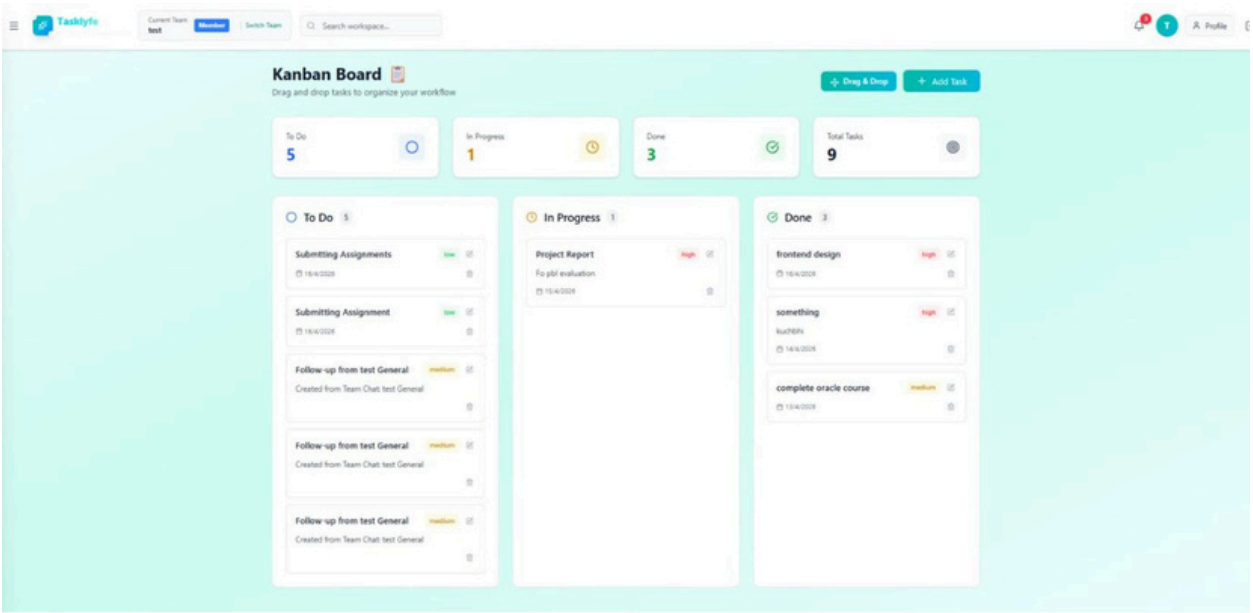

Fig-10 Kanban Board

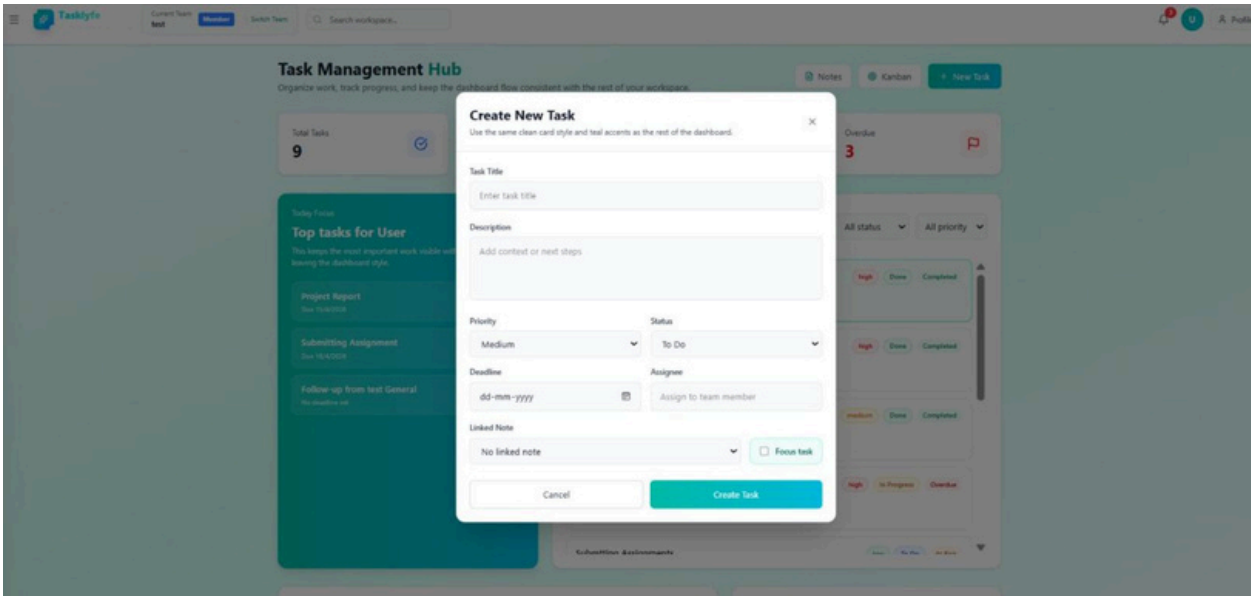

Fig-11 New Task Creation

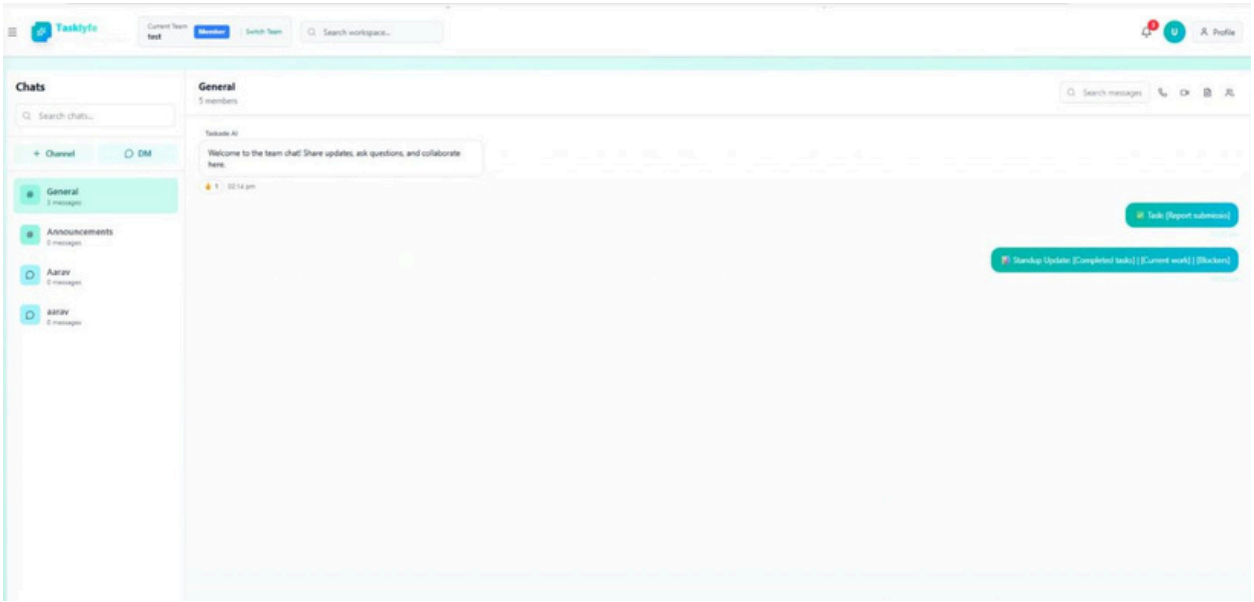

Fig-12 RealTime Chat

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